



SALAS O'BRIEN

3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project: Fern Parc; Richardson Housing Group
Location: Norcross, GA
Description: Evergreen TH - A, Slab, Right End Unit, Left-Garage

Date: October 1, 2025
By: JTH / TJH
House Faces: Worst West

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	9.0 ft
Floor Area	1,972 ft ²
Volume	18,518 ft ³
Percentage Glass	10 %
Number of Bedrooms	4

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	7193
SEER	15
ACH50	5.0
Mechanical Ventilation	NO
House Outside Air CFM	N/A

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.067	Basement	Concrete + 2x4	R-13 batt	
	W3	0.053	Attic	2x6	R-19 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.2	Exterior Door		R-5	
	D2	0.5	Garage Door		R-2	
Roofs	#	U-Value	Description/Notes			
	R1	0.026	Flat Ceiling	2x4	R-38 blown	
	R2	0.026	Vaulted	2x4	R-38 blown	
Floors	#	U-Value	Description/Notes			
	F1	0.053	Framed		R-19 batt	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.32	Blinds/Draperies	SHGC 0.22
	G2					
	G3					

LOAD SUMMARY					
			Unit 1 - Whole House		
			Cooling	Heating	
Sensible Load			24120 Btu/hr	24250 Btu/hr	
Minimum Nominal Capacity			2.68 Tons	7.11 kW	
Load Density			736 SF/Ton	3.6 W/SF	
Floor Area Served			1972.0 SF	1972.0 SF	
Mechanical Ventilation			N/A CFM	N/A CFM	
System Size			3.0 Tons	10.0 kW	
Leaving Air Temperature			57.2 deg F	96.3 deg F	
Installed Capacity			657 SF/Ton	5.1 W/SF	

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

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Unit 1 (Continued)				
Second Floor Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Stairs	21	35	28	
Hall	32	33	32	
Laundry	44	8	26	
Bath 2	14	17	15	
Bed 3	135	119	127	
Bed 3 WIC	47	47	47	
Bed 2 WIC	10	12	11	
Bed 2	101	111	106	
Bed 4	66	68	67	
Bed 4 WIC	5	4	5	
Primay WIC	25	46	35	
Primary Bath	37	63	50	
Primary Bed	134	101	117	
-	-	-	-	

COOLING LOAD CALCULATION					Ceiling Height		9		9		9		9		9		9		9		9		9		9		
Components			CLTD (°F)	U-Value (Btu/hr-sf-F)	Cooling Load Factor	Entire House		Foyer		Family		Dining		Kitchen		Pantry		Powder									
	Type	Description				Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr
Glass	G1	South	Right	(E)	19 Btu/hr-sf	62	1153		-	54	1,004	8	149		-		-		-		-		-		-		-
	G1	SouthWest		(SE)	29 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-
	G1	West	Front	(S)	37 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-
	G1	NorthWest		(SW)	26 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-
	G1	North	Left	(W)	7 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-
	G1	NorthEast		(NW)	21 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-
	G1	East	Back	(N)	32 Btu/hr-sf	37	1197		-		-	28.36	908	9	288		-		-		-		-		-		-
	G1	SouthEast		(NE)	26 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-
	G1	Sky Light			52 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-
	G1	Shaded			7 Btu/hr-sf	12	85		-		-		-		-		-		-		-		-		-		-
Door	D1	Door	21	0.20	4.20 Btu/hr-sf	16	69		-	12	50		-	4.452	19		-		-		-		-		-		-
	D2	Door	30	0.50	15.00 Btu/hr-sf	18	267		-	17.81	267		-		-		-		-		-		-		-		-
Above Grade Walls	W1	South	16	0.08	1.23 Btu/hr-sf	390	480	145	179	110	136	135	166	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1	SouthWest	21	0.08	1.62 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	W1	West	23	0.08	1.77 Btu/hr-sf	25	45	25	45		-		-		-		-		-		-		-		-	-	
	W1	NorthWest	19	0.08	1.46 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	W1	North	13	0.08	1.00 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	W1	NorthEast	19	0.08	1.46 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	W1	East	23	0.08	1.77 Btu/hr-sf	195	345		-		-	70	125	124	220		-		-		-		-		-	-	
	W1	SouthEast	21	0.08	1.62 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	W1	South	16	0.08	1.23 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	W1	SouthWest	21	0.08	1.62 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
Basement Walls	W1	West	23	0.08	1.77 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	W1	NorthWest	19	0.08	1.46 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	W1	North	13	0.08	1.00 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	W1	NorthEast	19	0.08	1.46 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	W1	East	23	0.08	1.77 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	W1	SouthEast	21	0.08	1.62 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	W2	Grade	0	0.07	0.00 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	W1	Partition	10	0.08	0.77 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	W1	Garage	30	0.08	2.31 Btu/hr-sf	313	723	125	290	150	347		-		-	38	87		-		-		-		-	-	
	W3	Attic	55	0.05	2.92 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
Floor	R3	Kneewall	55	0.05	2.92 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	W1	Ceiling	55	0.03	1.43 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
Roof	R2	Vaulted	55	0.03	1.43 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	F1	Crawl/Uncon.	11	0.05	0.58 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
Other	F1	Garage	30	0.05	1.59 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-	-	
	F2	Slab	0	0.10	0.00 Btu/hr-sf	745	0	71	-	294	-	150	-	181	-	20	-	29	-		-		-		-	-	
Internal	People				255 Btu/hr	1	255		-	1	255		-		-		-		-		-		-		-	-	
	Lights (W)				3.413 Btu/hr-W	374	1276	36	123	147	502	75	256	91	311	10	34	15	51	-	-	-	-	-	-	-	
	Appliances (W)				3.413 Btu/hr-W	930	3174		-	250	853		-	680	2,321		-		-	-	-	-	-	-	-	-	
	Infiltration	0.25	ACH		0.09 Btu/hr-cf		603		-	58		238		-	122		147		16		23		-		-	-	
Vent	Volume				CF	6705		639		2646		1350		1629		180		261		0		0		0		0	
	Area				SF	745		71		294		150		181		20		29									
Subtotal (Btu/hr)						9672		1096		3334		1744		3286		50		161		0		0		0		0	
Safety Factor						967																					
Sensible Cooling Load (Btu/hr)						10639																					
Nominal Tons Required (75% sens. cap.)						1.18																					
System Selection 1.5 Tons						600		68		207		108		204		3		10		-		-		-		-	
Space CFM																											
HEATING LOAD CALCULATION																											
Components																											
	Type	Description	ΔT (°F)	U-Value (Btu/hr-sf-F)	Heating Load Factor	Entire House		Foyer		Family		Dining		Kitchen		Pantry		Powder		0		0		0		0	
Glass	G1		54	0.32	17.3 Btu/hr-sf	111	1924	12	207	54	933	36.36	628	9	156	0	-	0	-	0	-	0	-	0	-	0	-
	G2		54	0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	G3		54	0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	D1		54	0.20	10.8 Btu/hr-sf	16	178	12	130	0	-	4.452	48	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	D2		54	0.50	27.0 Btu/hr-sf	18	481	17.81	481	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	W1		54	0.08	4.2 Btu/hr-sf	610	2535	170.2	708	110.2	458	205	852	124.3	517	0	-	0	-	0	-	0	-	0	-	0	-
Walls	W2		54	0.07	3.6 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	W3		54	0.05	2.9 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	W1	Partition	27	0.08	2.1 Btu/hr-sf	313	651	125.5	261	150	312	0	-	0	-	0	-	37.5	78	0	-	0	-	0	-	0	-
	W2	Grade	27	0.07	1.8 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
Roof	R1		54	0.03	1.4 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	R2		54	0.03	1.4 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	F1		54	0.05	2.9 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	F2	Slab	27	0.10	2.7 Btu/hr-sf	745	2012	71	192	294	794	150	405	181	489	20	54	29	78	0	-	0	-	0	-	0	-
Area					SF	745		71		294		150		181		20		29		0		0		0		0	
	Infiltration	0.25	ACH		0.24 Btu/hr	0	1629		155		643		328		396		44		63		-		-		-	-	-
	Subtotal (Btu/hr)						9409		2133		31																

Date: October 1, 2025
By: JTH / TJH
Level(s): Second Floor
SEER 15

Cooling Size	1.5	DX Cooling kWh	1678
Heating Size	1.5	RC Heating kWh	2321
Best Case	1.5	Total kWh	3999

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